

Beyond Anemia: Red Cell Distribution Width as an Emerging Biomarker in Ischemic Heart Disease and Acute Coronary Syndromes

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Context

Cardiovascular diseases remain a leading cause of morbidity and mortality worldwide. Their underlying patho-physiological mechanisms, including chronic inflammation, oxidative stress, and disturbances in iron metabolism, may impair erythropoiesis and alter erythrocyte indices. Among these, **Red Cell Distribution Width (RDW)** has gained increasing attention as a potential prognostic biomarkere

Objectif

This study aimed to characterize the RDW profile in patients with cardiovascular diseases and to investigate its association with clinical and biological characteristics, as well as with different types of cardiac disorders

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A cross-sectional observational, descriptive, and analytical study was conducted between February and May 2026 in the Cardiology Department of Saadna Abdenmour University Hospital, Sétif, Algeria. Sociodemographic, clinical, and biological data were prospectively collected from patients' medical records. Participants were stratified into two groups according to RDW-CV values: normal (11.5–15.0%) and elevated ($\geq 15.0\%$). Statistical analyses were performed using SPSS® Statistics version 26.0, with statistical significance set at $p < 0.05$.

Results

Elevated RDW was significantly associated with acute coronary syndrome ($\chi^2 = 13.857$; $p = 0.001$; Cramer's $V = 0.462$) and ischemic heart disease ($\chi^2 = 6.027$; $p = 0.014$).

Result

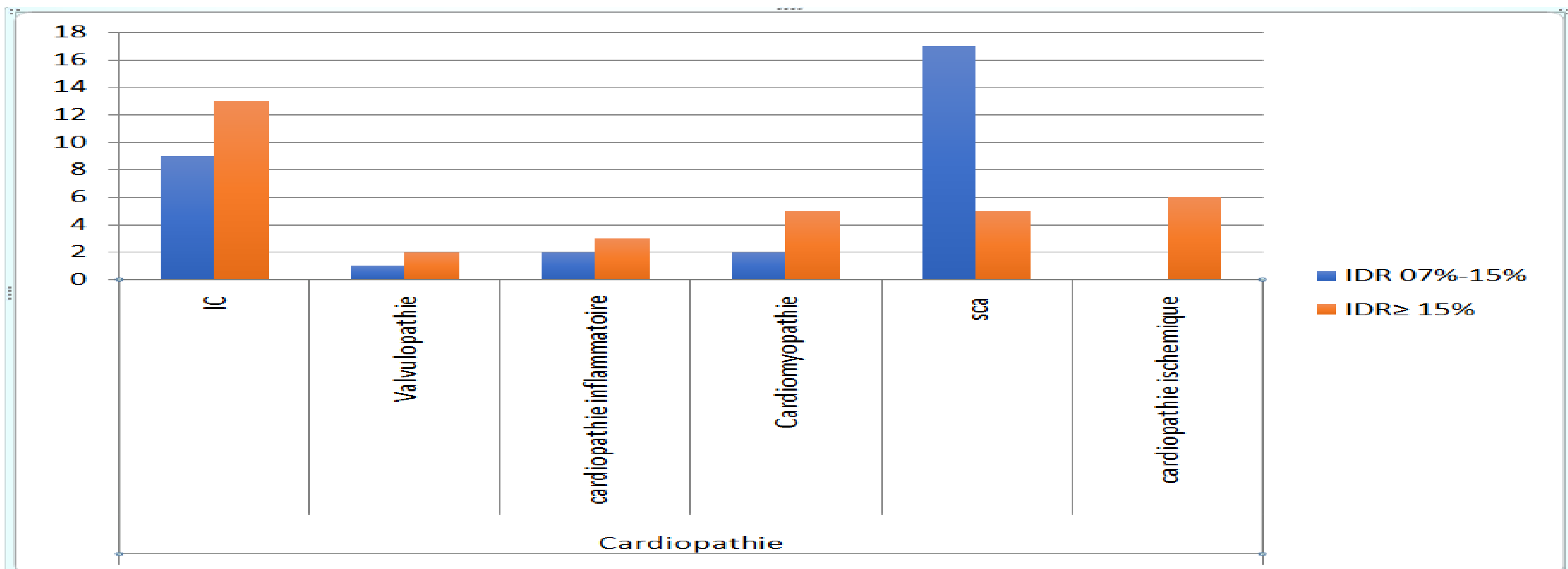


Figure 1: répartition de l'IDR sur les cardiopathies

The analysis showed significant variation in RDW according to the type of heart disease. An RDW $\geq 15\%$ was particularly frequent in ischemic heart disease (100%), cardiomyopathy (71.4%), valvular heart disease (66.7%), and heart failure (59.1%), whereas acute coronary syndrome (ACS) was mainly associated with an RDW between 7% and 15% (77.3%).

Fisher's exact test with Monte Carlo simulation confirmed a significant association between elevated RDW and the type of heart disease ($p = 0.004$). Similarly, the Kruskal–Wallis test showed a significant difference in RDW values across the different types of heart disease ($H = 15.760$; $p = 0.008$), suggesting that the type of heart disease influences the distribution of RDW.

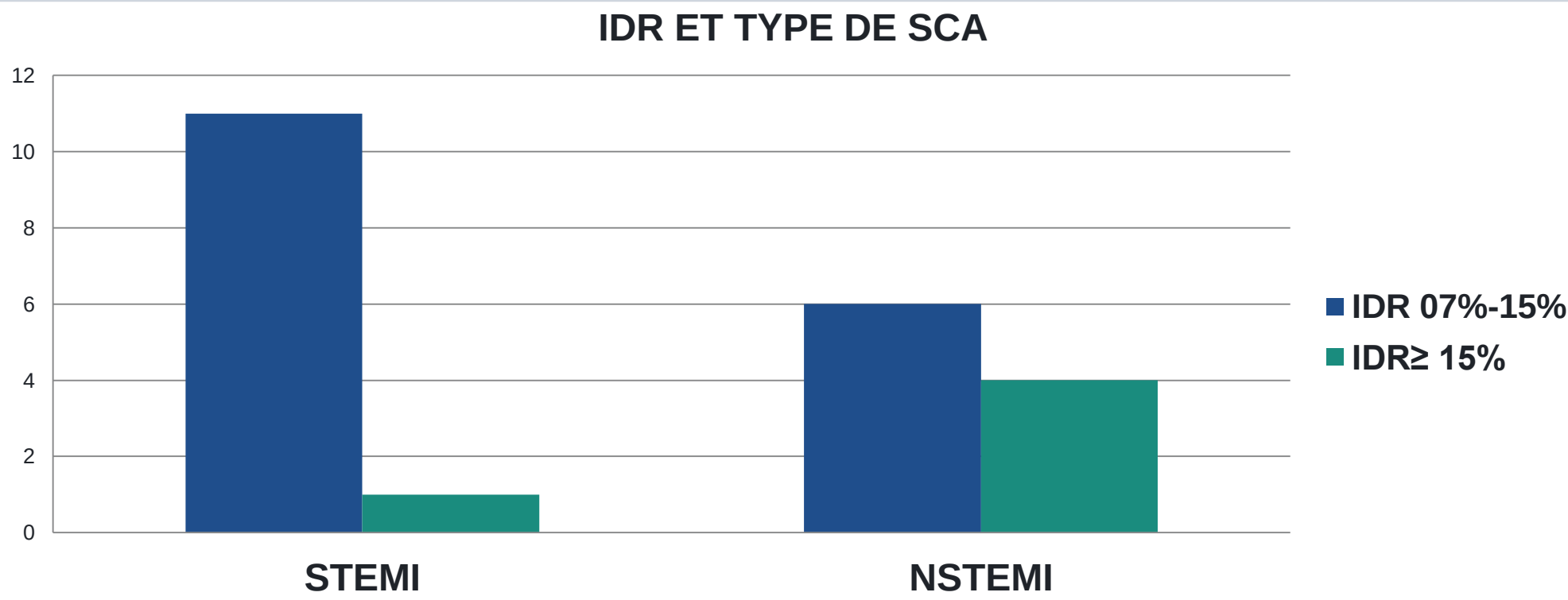


Figure 2: répartition de l'IDR selon le type de SCA

Sample 1-Sample 2	Statistiques de test	Erreur standard	Statistiques de test standard	Sig.	Sig. Sig. ^a
STEMI-NSTEMI	-13,292	8,090	-1,643	0,100	0,301
STEMI-non	-23,047	6,168	-3,736	0,000	0,001
NSTEMI-non	-9,756	6,633	-1,471	0,141	0,424

Tableau 1: Comparaisons appariées de Type SCA

The analysis showed a significant association between RDW categories and the type of ACS ($\chi^2 = 13.857$; $p = 0.001$), with a moderate strength of association (Cramer's $V = 0.462$). Elevated RDW was particularly associated with NSTEMI, suggesting a link with chronicity and biological vulnerability. STEMI patients were predominantly characterized by intermediate RDW values. The Kruskal–Wallis test also confirmed a significant difference in RDW values according to ACS type ($H = 14.491$; $p = 0.001$). This difference was mainly observed between STEMI and patients without ACS ($p = 0.001$), with no significant difference between STEMI and NSTEMI or between NSTEMI and non-ACS patients.

Discussion

The study conducted at Sétif University Hospital in 65 patients showed that the **red cell distribution width (RDW)** varied significantly according to the type of heart disease ($p = 0.008$), with elevated RDW observed in 100% of patients with **chronic ischemic heart disease** ($p = 0.014$). A significant association was also observed with **acute coronary syndrome**, particularly with **NSTEMI** (40% of patients with RDW $\geq 15\%$ compared with 8.3% among STEMI patients; $p = 0.001$).

Conclusion

RDW is a simple, readily available, and cost-effective hematological biomarker associated with ischemic heart disease, particularly acute coronary syndrome. Its integration into the biological assessment of patients may improve cardiovascular risk stratification. Nevertheless, large-scale multicenter studies are required to validate these findings and establish the optimal clinical decision threshold for RDW.